40th Fluid Dynamics Conference and Exhibit
28 June - 1 July 2010, Chicago, Illinois

AIAA 2010-5112

LES Study of Feedback-loop Mechanism of Supersonic Open Cavity Flows

Weipeng Li[1]
Dept. of Aeronautics and Astronautics, University of Tokyo, Tokyo, 113-8656, Japan

and

Taku Nonomura[2] Akira Oyama[3] and Kozo Fujii[4]
Institute of Space and Astronautics Science/JAXA, Sagamihara, Kanagawa, 252-5210, Japan

Supersonic flow over a three-dimensional rectangular cavity with length-to-depth ratio of 2 is numerically studied by implicit large-eddy simulation to clarify the feedback-loop mechanism. A feedback-loop cycle is described and visualized with phase-averaged analysis of simulation results. Causality between the feedback acoustic wave and leading-edge shedding vortex is clearly demonstrated. Mach wave reflection at trailing edge is turned out to be the generation mechanism of feedback acoustic wave. It is convinced by investigating time-series instantaneous flowfields and auto-correlation coefficients of three simulation cases with different convective Mach number. Components of compression waves in supersonic cavity flows are summarized and their features are discussed. Proper orthogonal Decomposition (POD) in frequency domain is firstly employed to analyze wave propagations inside cavity. Results statistically show the propagation traces of notable compression waves inside cavity which are affected by high-speed recirculation flows.

I. Introduction

HIGH-speed flows over cavities have been studied for many years both for the interests of practical perspectives and natural physics of fundamental fluid dynamics. The utilization of supersonic cavities in aerospace engineering includes internal store carriage on high-performance aircrafts, flame-holder for scramjet engines and flow control for supersonic nozzles and jets.[1-5] The fluid-dynamic regime of supersonic cavity flows involves shear-layer instability, vortices impinging, acoustic radiation, and acoustic/shear-layer interactions. Flow over cavities can be categorized as open, closed and transitional cavity flows.[1,6] In usual, open cavity flows occur at length-to-depth (*L/D*) ratio less than 10 under high-speed inflow conditions when the shear-layer reattachment takes place near the aft wall. One of distinguishing features of open cavities is the presence of discrete and intense cavity tones, especially for high-speed cavity flows, the sound pressure level of around 160 dB have been observed for a cavity under Mach 2.0 flow.[7,8] Such high amplitude noise could lead to some harmful effects, such as structural fatigue and damage, adverse affects on store separation and undesirable noise. Thus, the generation mechanism of cavity tones needs to be clarified for effective control and suppression approaches design.

The driving mechanism of cavity tones is well recognized as the feedback loop between shear-layer instability and acoustic forcing. It was first put forward by Powell[9] on the study of edge tones and could be found in many aerodynamic applications, such as jet screech, cavity tones and jet impingement. In 1964, Rossiter[10] put forward a semi-empirical formula for prediction of cavity tones based on experimental data, which was widely used as benchmark for numerical simulations both at subsonic and supersonic conditions when the experimental data are unavailable. Heller and Bliss[11] modified the formula by taking accounts sound speed inside cavity was related with recovery temperature of the flow. In simplicity, the feedback-loop mechanism could be described as three indispensable steps: (1) generation of feedback noise related to the motion of large-scale structures at trailing edge

[1] Graduate Student, Department of Aeronautics and Astronautics, University of Tokyo, AIAA Student Member
[2] Postdoctoral fellow, Department of Space Transportation Engineering, 3-1-1 Yoshinodai, AIAA Member.
[3] Associate professor, Department of Space Transportation Engineering, 3-1-1 Yoshinodai, AIAA Member.
[4] Professor, Department of Space Transportation Engineering, 3-1-1 Yoshinodai, AIAA Fellow.

Copyright © 2010 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved.

(2) propagation of feedback acoustic wave to upstream (3) excitation of unstable shear-layer at leading edge, which leads to further vortex shedding and large-scale structures.

Although the feedback-loop mechanism had been well-established and accepted, some physical concepts of feedback-loop mechanism are still not well-understood, especially for high-speed compressible cavity flows. In Zhuang's experimental work[7], the pressure spectrum cannot match well with the prediction of Rossiter's model. Ünalmis[12] indicated that the cavity oscillation frequencies are accurately predicted using simple closed-box acoustic theory. In Vikramaditya's work[13], the presence of a forward-moving acoustic wave could not be identified in the cavities with small ramp angles while the reasons for this behavior still had not been understood. In early years, the mechanism proposed by Heller and Bliss[14] supported that the aft wall mass injection and removal was responsible for the initially generation of internal cavity waves. However, to the authors' knowledge, two key processes of feedback-loop mechanism still had not been verified yet. First, the direct relation between feedback acoustic waves and shedding vortex at leading lip had not been clearly visualized both in experimental and numerical studies. Second, the generation mechanism of feedback acoustic waves is not clear yet, especially for high-speed cavity flows involving shock waves, Mach waves and high-speed convective shear-layer. Thus, many open questions are still remaining underlying the basic physical phenomenon and further investigations should be conducted.

In order to clarify the feedback-loop mechanism, many significant efforts had been achieved both in experimental studies and numerical studies. Overviews of the previous works on cavity flows could refer to the investigation of Grace[1], Colonius[2], Rockwell[15], and Lawson[16]. Thus far, experimental studies had been conducted by many researches, e.g. Rossiter[10], Heler and Bliss[11], and Zhuang and Alvi.[7,8] However to-date very limited data are available for supersonic cavity flows due to inherent difficulty in measurement, particularly in terms of detailed flowfield data.[8] Numerical simulation is an alternative way to provide insight into physics of supersonic open cavities and had been conducted for many years.[17-24] Majority of these studies were Reynolds averaged Navier-Strokes (RANS) simulation with various turbulence models. Tam[17] solved time accurate thin layer Navier-Stokes equations for unsteady supersonic open cavity flows and described the self-sustained oscillation cycle. He found that the pressure pulses at trailing edge had the same frequency as vortex shedding. Shieh[18] adopted two-equation turbulence model for the two-dimensional unsteady supersonic open cavity flow to analyze physical mechanism envisioned by the experiments. Zhang[19] analyzed the flow-acoustic oscillation of a supersonic cavity using k-ω turbulent model. Using unsteady RANS coupling k-ω turbulence model, Aradag[20] calculated three-dimensional supersonic open rectangular cavity and investigated three-dimensional features of the self-sustained oscillations.

However, RANS simulations are generally dissipative to capture weak pressure fluctuations and high-frequency tones. Accurate simulation of the radiated acoustic waves together with the unsteady turbulent shear-layer is needed to capture small amplitude of the acoustic field and their propagations over a long distance with little attenuation of numerical dissipation. Recently, large-eddy simulation (LES), detached-eddy simulation (DES), and RANS/LES hybrid methods were preformed for supersonic cavities simulation with low-dispersive and low-dissipative numerical schemes. Rizzetta[21] conducted LES studies using a fourth-order compact difference scheme to investigate the efficiency and mechanism of the active noise suppression with pulsed mass injections. In Hamed's study,[22] DES are performed for unsteady three-dimensional turbulent supersonic open cavity flow and results revealed the flow features of supersonic cavity flow, including the vortex shedding, shock waves, and coupling of the acoustic and vorticity fields. From the results of hybrid RANS/LES simulations, Arunajatesan[23] analyzed the control mechanism and dynamic loading of cavity tones using a rod spoiler as a device of passive actuation. Using fifth-order weighed essentially non-oscillation(WENO) scheme, Liang[24] studied two-dimensional supersonic cavity flows with hybrid RANS/LES method to analyze the effects of inflow conditions and geometry structures on self-sustained oscillation characteristics. In all numerical studies mentioned above, the generation mechanism of cavity tones in supersonic cavity flow was not well answered, and detailed discussions of each component of compression wave were not investigated, which is important for us to clarify the mechanism of cavity tones and to suppress cavity tones.

The aim of the present study is to clarify feedback-loop mechanism and analyze each notable compression wave in supersonic cavity flows. Large-eddy simulations (LES) with high-resolution schemes are performed for a three-dimensional rectangular supersonic open cavity (L/D=2) under the condition of freestream Mach number of 2.0 and Reynolds numbers of 10^5, respectively. Phase-averaged analysis will be used to analyze feedback loop to reveal the causality between the feedback acoustic wave and shedding vortex at the leading edge. Also the generation mechanism of feedback acoustic wave will be investigated by visualization of instantaneous flowfields of three cases with different connective Mach number. Components and features of compression waves in supersonic cavity flow will be summarized. Due to complex waves inside cavity and their nonlinear interactions, Proper orthogonal Decomposition (POD) in frequency domain will be firstly employed to analyze the propagation of notable compression waves.

II. Numerical methods

2.1 Computational Conditions

Supersonic flow pass over a cavity of L/D=2, W/D=1 are numerically studied, where L is the streamwise length of cavity, W is spanwise length of cavity and D is the depth of cavity. Freestrem Mach number is set to 2.0 and the Reynolds number based on cavity depth is set to 10^5. Laminar velocity distribution is profiled for the inflow condition with initial momentum thickness of shear-layer fixing to 0.01D at the leading lip.

2.2 Governing equations and algorithm

The governing equations are three-dimensional compressible Navier-Strokes equations. Implicit large eddy simulation (ILES) is conducted, which relies upon the numerical scheme to dissipate turbulent energy. In order to meet the requirements of low dispersion and dissipation for computational aeroacoustic(CAA) and high-resolution of turbulence flows, the spatial derivatives are computed by the 7th-order weighted nonlinear compact scheme (WCNS)[25,26] which had been validated effective for high-speed jet flows and aeroacoustic simulations[27]. The numerical fluxes are evaluated by the simple high-resolution upwind scheme(SHUS)[28] which is a family of advection-upstream-splitting-method(AUSM) type schemes. Viscous terms are evaluated by the sixth-order difference scheme. Alternate directional implicit symmetric Gauss-Seidel (ADI-SGS) scheme[29] is applied for time integration and Newton-like sub-iteration is used and 2nd order temporal accuracy is maintained. The maximum CFL number is approximately equal to 1.2.

2.3 Grid and boundary conditions

The side-view of computational domain is showed in Fig. 1. The computational domain is composed of inside cavity and upper cavity region. Structural grids are used. Grid points are 200x150x80 for the inside cavity region, and 360x150x80 for the upper cavity region. Total grid number is approximate 6.7 millions. The length from inflow boundary to the cavity leading-edge lip is equal to cavity depth, and 4D is extended from trailing-edge lip to outflow boundary. A distance of 4D is extended in the vertical direction and no buffer region due to supersonic freestream. In the spanwise direction, 80 points are equally distributed within the total length of 1D. The grid spacing in refined at near wall region to sufficiently resolve behavior of shear-layer. The values of ΔX^+, ΔY^+, ΔZ^+ are 5.0, 1.0 and 12.5 respectively at the cavity leading edge. In order to validate the grid resolution, a finer mesh which is mainly refined inside cavity region is also conducted. The total number of grid points is 18.3 millions.

No-slip adiabatic wall boundary condition is imposed at all the wall boundaries. Zero-gradient pressure condition is employed at the outflow and Blasius laminar profile is adopted at the inflow boundary. Periodical boundary condition is imposed in the spanwise direction.

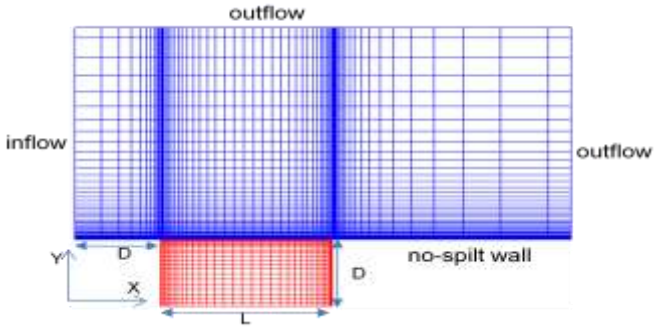


Fig. 1 Computational grid (shown every other ten points)

III. Validations

The accuracy of our simulations is validated in two ways, investigating grid convergence and comparing with experimental data in Zhuang's studies.[7,8] The distributions of spanwise-averaged time-mean pressure coefficients along internal cavity walls appears in Fig. 2, which shows the two different computational grids have almost the same distributions.

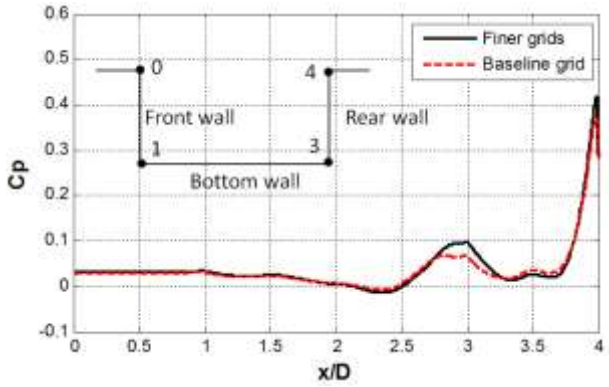


Fig. 2 Spanwise-averaged time-mean pressure coefficient distributions



Fig. 3 Streamline of internal cavity

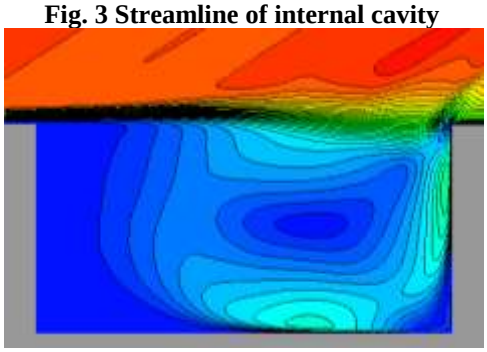


Fig. 4 Mach number contours of spanwise-averaged time-mean flowfield

The streamline of internal cavity is presented in Fig. 3 and similar distribution could be found in the result of finer mesh. It has large recirculation region closed to rear wall due to high-speed mass injection near trailing edge. In the left bottom corner, a smaller reverse recirculation region exists. Mach number contours of spanwise-averaged time-mean flowfield are showed in Fig. 4. The Mach number of recirculation flow is up to approximately 0.8 near rear wall and 0.6 near bottom wall. Such high-speed reverse flow roughly corresponds to the observation in Zhuang's experimental work, in which reverse flow as high as 40-50% of free-stream speed was reported.

The pressure spectrum is shown in Fig. 5 with comparison with experimental result conducted by Zhuang et al.[8] Result shows that the main modes of cavity tones observed by baseline grids simulations agree with that of finer grids simulation result. The peak values of pressure spectrum in low-frequency agree with Zhuang's experimental results. Although disagreements occur at high-frequency modes, which might be caused by non-linear interaction among different modes, they do not affect our discussion on feedback-loop mechanism of primary modes. Note that the inflow fields of experimental result are turbulent and Reynolds number is slightly higher than our simulation.

The capacity of noise capture in our simulation is demonstrated in Session VI with the analysis of noise components in supersonic cavity flows. Major compression waves are captured, which also agree with experimental observation.

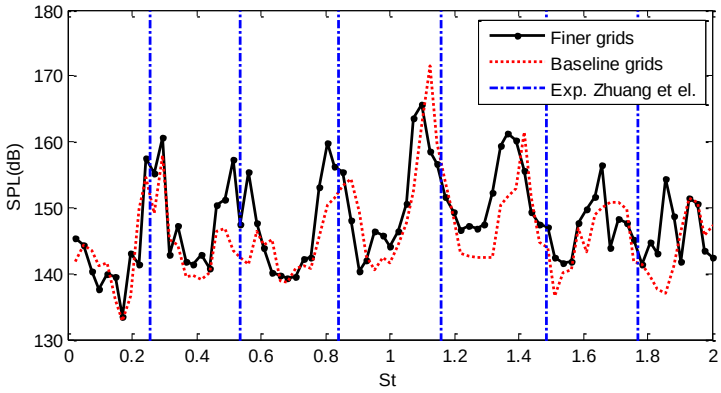


Fig. 5 Pressure spectrum

IV. Feedback Mechanism

Phase average analysis is performed to analyze the self-sustained oscillation and visualize the feedback-loop cycle. This technique[30] decompose a flow variable $f(t)$, into three components as a time(ensemble)-averaged mean component $\bar{f}$, a phase-varying component $\tilde{f}$, and a random component f'. The phase-averaged component is then defined as $f(t) = \bar{f} + \tilde{f} + f'$. Larchevêque et al.[31] had used this technique on subsonic deep cavity flows to detect coherent structures. Before phase definition, low-pass filtering is implemented on the signals of spanwise-averaged density fluctuations at the mid of cavity front wall. Fig. 6 shows the time-series signals of original density fluctuations and low-pass filtered ones. The lowest values(black dots in Fig. 6) of the low-pass filtered signals are defined as the trigger points for phase averaging. Phase number is set to 20 in this study.

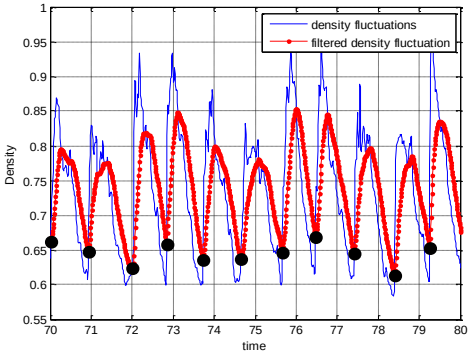


Fig. 6 Low-pass filtered spanwise-averaged density fluctuations

The feedback loop is visualized in Fig. 7 using phase-averaged analysis for spanwise-averaged data. The black-line contours is the positive values of the second invariant of velocity gradient tensors ($Q_{2nd} > 2(U_\infty/L)^2$), which represent the rolling vortices. The background color counters is density gradient in streamwise direction by which compression waves are clearly observed and the forward and backward traveling compression waves inside cavity can be distinguished. The feedback loop is described as bellows.

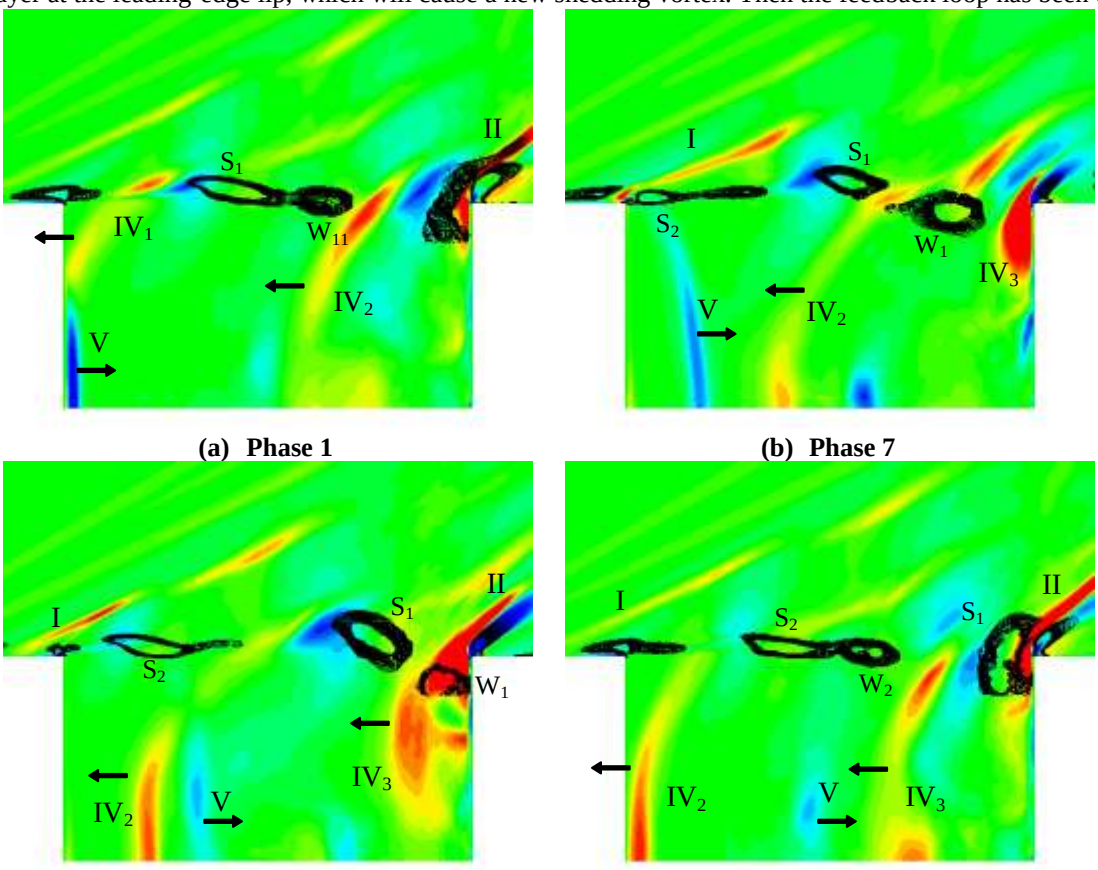
1) The forward-traveling feedback compression wave (IV_1) reached the front wall and part of the wave had been reflected off the front wall and generated backward-traveling reflection wave(V). The upper part of feedback compression wave travels associated with the inside part. At the same time, another feedback compression wave (IV_2) is propagating toward front wall. Two large-structural vortices(S_1 and W_1) are observed in center part of cavity and convecting toward trailing edge. The front vortex(W_1) was generated by Kelvin-Helmholtz instability. The following vortex(S_1) was generated by acoustic disturbance at leading edge and strengthen by Kelvin-Helmholtz instability. Large structural vortices impinged on the rear wall and high amplitude of density gradient exists at the trailing edge.
2) The feedback compression wave (IV_1) reached the leading-edge lip and induced disturbance to the shear-layer. A new shedding vortex(S_2) was generated by the acoustic forcing with little time delay. The compression wave(V) is the front wall reflection wave of feedback compression wave(IV_1). The other feedback compression wave(IV_2) continue propagating toward front wall. The vortex(W_1) generated by

Kelvin-Helmholtz instability injects into cavity and convects toward rear wall. A new intense feedback compression wave(IV_3) is generated at trailing edge and propagates upstream. Based on our simulation the generation mechanism of feedback compression wave is less related to vortices impingement rather than reflection of Mach wave following the large-scaled shedding vortex. This phenomenon will be discussed in detail later.

3) The front wall reflection wave(V) and forward-traveling feedback compression wave(IV_2) met inside cavity and propagate in their directions, respectively. The vortex(W_1) impinge on rear wall, and two shedding vortexs(S_1 and S_2) behave actively in shear-layer. New generated feedback compression wave(IV_3) continue propagating toward front wall. Compression wave I and II are shock waves at leading and trailing edge respectively, which will be discussed in Session VI.

4) The shedding vortex(S_1) impinged on trailing-edge lip and was split into two parts. One part convects to downstream, and the other part injects into cavity and cause large mass injections. New votex(W_2) was generated by Kelvin-Helmholtz instability. The front wall reflection compression wave(V) propagate toward rear wall and its propagation direction will be affected by recirculation flow. From the visualization, the front wall reflection wave(V) becomes weak as it propagate toward rear wall and does not reach rear wall with high amplitude.

As time goes by, the forward-traveling feedback compression waves(IV_2) will reach the front wall and excite the shear-layer at the leading-edge lip, which will cause a new shedding vortex. Then the feedback loop has been closed.



(a) Phase 1 **(b) Phase 7**

(c) Phase 13 **(d) Phase 19**

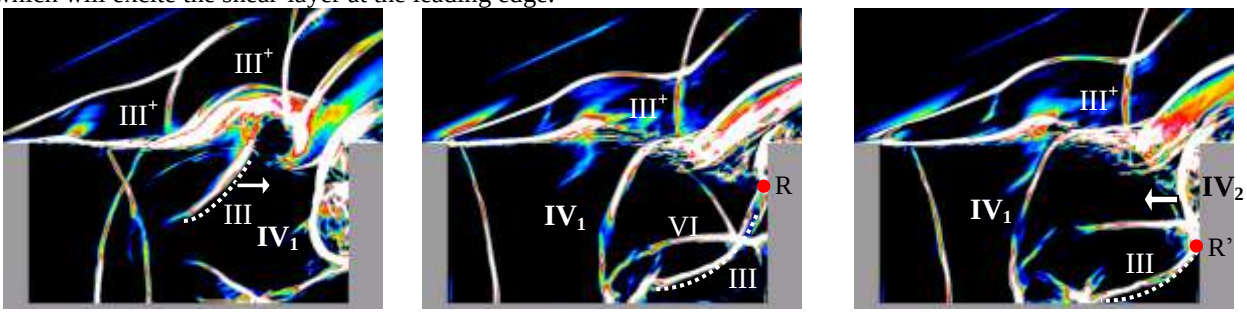
Fig. 7 Feedback loop visualization using phase-averaged data

From the feedback loop visualization, two points will be pointed out for the understanding of the mechanism of fluid-acoustics feedback-loop. First, the shedding vortex at leading edge is induced by acoustic forcing and the time delay between the acoustic disturbance and vortex shedding is very small. The other is that Mach wave reflection near rear wall seems responsible for the generation of feedback acoustic wave. The vortices impingement and mass injection may weakly contribute to the generation of feedback acoustic wave. The following section presents detailed discussion on Mach wave reflection.

V. Mach wave reflection

Feedback acoustic wave is the crucial factor in self-sustained oscillation. It is widely accepted that the feedback acoustic wave is generated by the vortices impinging on the trailing edge and a time-delay exists between the vortices impinging and the acoustic radiation. However, from our simulation visualization, the feedback acoustic is much more related to Mach wave reflection.

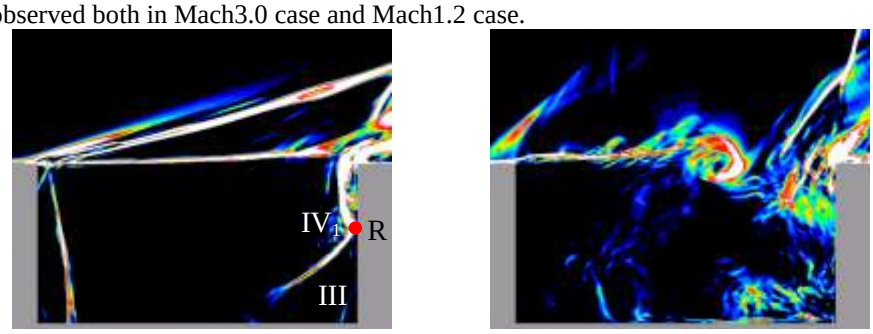
The absolute density gradient contours of instantaneous spanwise-averaged flowfields are shown in Fig 8. Compression wave III in Fig. 8.a indicates the Mach wave generated by the high-speed convective shear-layer. Compression wave IV_1 indicates a feedback acoustic wave. As the wave propagating, Mach wave III reached the rear wall and the leading part of Mach wave III is reflect by the rear wall as showed in Fig. 8.b. Point R in Fig. 8.a is the reflection point. The reflection continues as Mach wave propagating toward to rear wall. In Fig. 8.c, large part of Mach wave was reflected by rear wall and reflection point goes down to point R' in Fig. 8.c. A new feedback acoustic wave IV_2 is generated. The reflection eventually results in the forward-traveling feedback acoustic wave, which will excite the shear-layer at the leading edge.



(a) t=τ0 **(b) t=τ0+0.6** **(c) t=τ0+0.7**

Fig. 8 Absolute density gradient contours of instantaneous spanwise-averaged flowfields (Ma 2.0)

In order to convince our conclusion of Mach wave reflection, two other case are conducted under free-stream Mach number of 1.2 and 3.0 with same initial inflow shear-layer thickness, which represent lower and higher convective Mach number of shear-layer. As we know, Mach waves radiate when the difference between the freestream velocity and the convection velocity of the large-scale structures exceeds the local speed of sound. If the convective Mach number is subsonic, there is no intense Mach wave generation. So the comparison among these three cases could result in the relationship between feedback acoustic waves and Mach waves. The instantaneous flows are shown in Fig 9. Fig. 9.a clearly shows the Mach wave III and generated feedback acoustic wave IV in the Mach3.0 case. Point R indicates the reflection location. The simple components of compression waves are related to relatively stable shear-layer due to compressibility effect which leads to smaller spreading rate. The visualization of instantaneous flowfiled of Ma1.2 case is showed in Fig. 9.b. There is no intense feedback acoustic wave inside cavity although with the presence of large-scale vortices. The absence of intense feedback acoustic wave probably results from no Mach waves generated by unstable shear-layer at subsonic convective Mach number. To be notable, cavity tones are observed both in Mach3.0 case and Mach1.2 case.



(a) Ma 3.0 **(b) Ma 1.2**

Fig. 9 Absolute density gradient contours of instantaneous spanwise-averaged flowfields with different freestream Mach number

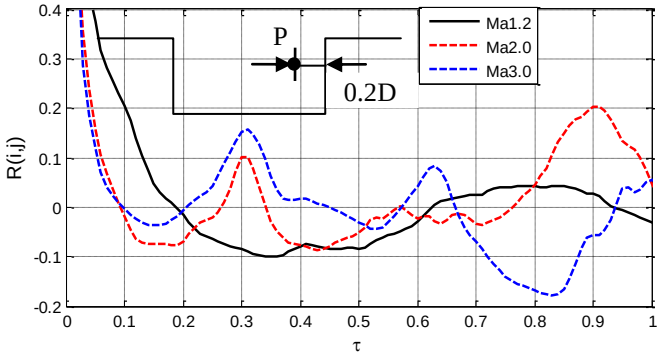


Fig. 10 Auto-correlation coefficients of three simulation cases

Fig. 10 shows auto-correlation coefficients of three cases at point P, where has a distance of 0.2D from rear wall. The values of absolute density gradient are used for calculation auto-correlation coefficients in the purpose of detecting compression waves. For Mach1.2 case and Mach 3.0 case, a peak value is observed with delay time of approximately 0.3, which indicates the time between Mach wave reaching point P and its reflection wave reaching point P. The delay time is less than 0.4 because the local sound speed near trailing edge is large than 1.0 and the propagation of Mach wave is inclined to rear wall. However, for the Mach1.2 case with subsonic convective Mach number, no peak value is observed within delay time of 0.4. This could be explained by the absence of Mach wave for subsonic convective shear-layer.

From the visualization of three different Mach number conditions, we could draw the conclusion that the feedback acoustic wave is generated by Mach wave reflection for the cases with supersonic convective shear-layer. However, following questions are raised up: 1) how the intense Mach wave is generated, and 2) whether the Mach wave reflection is common in any open supersonic cavity flows. In our simulations, the inflow filed condition is profiled in laminar distribution. When the shear-layer at the leading lip is excited by acoustic disturbance, highly two-dimensional vortex would be shed from leading lip. It will grow into large-scale structure with highly two-dimensional characteristic and large shear-layer deforming as convecting downstream. Due to highly tow-dimensional shear-layer behavior, both sides of shear-layer will radiate intense Mach waves. The amplitude of Mach waves might be weaker if the shear-layer fluctuations are more three-dimensional. Further studies will be conducted to clarify the mechanism under turbulent inflow conditions.

VI. Components of compression waves in supersonic cavity flows

Different from the subsonic cavity flows, more complex compression waves are present in supersonic cavity flows. The components of compression wave of supersonic cavity flows and their features are summarized in present study. Based on post-processing visualizations of the simulation results, the compression waves in supersonic open cavity flows are categorized as five components: (1) leading-edge shock waves; (2) trailing-edge shock waves; (3) Mach waves; (4) feedback acoustic waves; (5) reflection waves.

Intermittent compression and expansion waves radiate from leading edge because of shear layer flapping at the leading edge. We call the compression wave as leading-edge shock wave, which was showed as compression wave I indicated in Fig. 7. Fig.11 shows the pressure contours of time-mean flowfields which indicates the intensity of the leading-edge shock waves. However, the expansion waves radiated from leading edge are not observed in our simulation. Fig. 11 shows high amplitude of pressure distribution is observed near trailing edge, which is caused by the unsteady trailing-edge shock waves. Due to large shear-layer deformation and impingement of large coherent structures, intense shock waves are periodically generated near trailing edge, which is shown as compression wave II in Fig. 7.

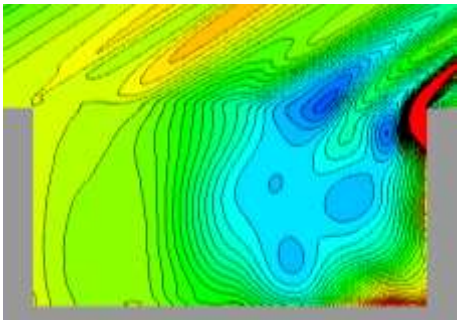
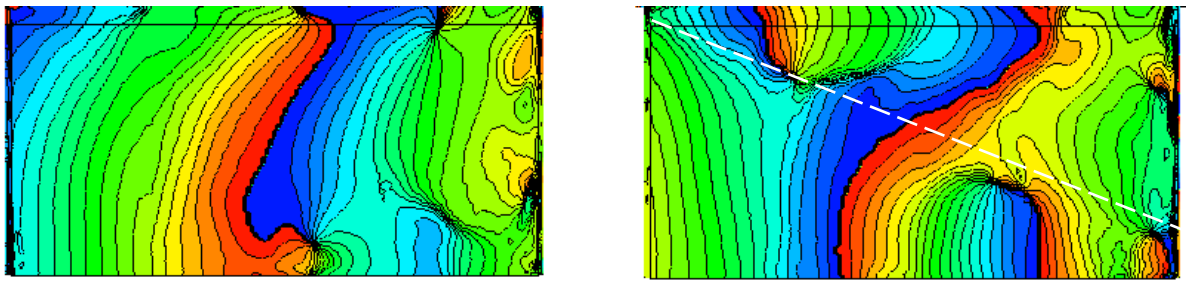


Fig. 11 Pressure contours of time-mean flowfields

Mach wave is a distinguishing feature of supersonic cavity flows. Both sides of shear-layer could radiate Mach waves. Mach wave in the upper region, which is showed as III^{+} in Fig. 8, periodically radiate from wavy-like shear-layer and propagation downstream. Inside cavity, Mach waves are also generated and it is responsible for the intense feedback acoustic wave as discussed in Session 5. The feedback acoustic wave is the key to reveal the mechanism of fluid-acoustic resonances. It propagates forward and will be reflected by the front wall of cavity. The front wall reflection compression wave is shown as V in Fig. 7. Front wall reflection wave is very important if researchers attempt to demonstrate the cavity tones are derived by purely acoustic modes or closed-box theory. Apart from the front wall reflection wave, the feedback compression wave could also be reflected by the bottom wall. As visualized as VI in Fig. 8, clear bottom wall reflection waves is observed.

The compression waves inside cavity, mainly including Mach waves, feedback acoustic waves and front wall reflection waves, are complex and their propagations are difficult to analyze owning to the interactions among different waves and high-speed recirculation flows. In present study, Proper orthogonal decomposition (POD) is used to analyze wave propagation inside cavity. Proper orthogonal decomposition (POD), also known as principal component analysis, is a useful and prevalent data-mining technique to extract low-dimensional descriptions with capture major features from large amounts of high-dimensional data, which had been used for analysis of turbulent flow, plane mixing layers, and jet noise.[32-34] Freund et al.[35] implemented POD analysis to obtain low-order models to accurately capture the oscillation. However, no researcher attempt to use POD to capture the acoustic propagation in the supersonic cavity flows. The conventional snapshot POD method is incapable to remain the phase information of periodical oscillation. Fourier transform decompose flowfields into amplitude parts and phase parts, while the phase parts contain the time-delay information represent wave propagation features. Then the POD analysis in frequency domain could get the major energetic mode of phase information. The positive and negative values of density gradient in streamwise direction are separately chosen as characteristic parameter to distinguish the forward and backward-traveling compression waves. Strouhal number used for Fourier transform is selected as the maximum peak value in pressure spectrum. Snapshot technique is employed to extract the most energetic mode, where regarding Fourier-transformed data as snapshots with overlapping operation.



(a) Mode one with 62.7% energy **(b) Mode one with 85.4% energy**

Fig. 12 Contours of phase part by POD analysis in frequency domain using data of (a) positive streamwise density gradient; (b) negative streamwise density gradient

The phase distributions of POD analysis are showed in Fig. 12. The forward propagating waves, mainly related to feedback acoustic waves, are described in Fig. 12.a, which is the result of POD analysis using positive values of streamwise density gradient. The propagation traces can be readily obtained by the distributions of the contour lines. The feedback acoustic waves start from the region near the trailing edge, and quickly reach the bottom wall. The propagation trace become like arc after short time propagation, and the inclined angles from streamwise direction are large than 90 near the bottom wall. Some disturbance distributions are observed near wall region which probably

caused by the violent vortices behavior and intense reflection waves. When the feedback acoustic waves pass approximately half of cavity, the propagation seems regular and the inclined angles from steamwise direction are approximately vertical to bottom wall. The change of propagation direction is mainly related to high-speed recirculation flows. Near the rear wall and downstream half of bottom wall the velocity of recirculation flows is very high as showed in Fig.4. The speed of wave propagation in these two regions is the sum of local sound speed and recirculation speed. Then the inclined angles are changed near bottom wall. The propagations of Mach waves and front wall reflection waves are visualized in Fig. 12.b. The contour lines in the region below white dash line show the propagation of front wall reflection waves. It is also affected by the high-speed recirculation flows. In the upper region from white dash line, Mach waves start near trailing edge and propagate towards rear wall. The radiation angles of Mach wave are related to gradients of the contour lines. POD analysis in frequency domain is an efficient technique to analyze wave propagation. Streamwise density gradient is an appropriately parameter to separate the forward-propagating and backward-propagating waves inside cavity using negative and positive values respectively. Results clearly show the propagation traces of notable waves inside cavity and high-speed circulation flows influence the propagation.

VII. Conclusions

Large-eddy simulations(LES) with high-performance turbulence/aeroacoustic resolution methods were performed on a three-dimensional supersonic rectangular open cavity (L/D=2) under freestream Mach of 2.0 and Reynolds numbers of 10^5 respectively. Feedback loop is described and visualized using phase-averaged analysis. The causality between the feedback acoustic wave and shedding vortex at the leading edge is clearly showed. The feedback acoustic wave was found be generated by the reflection of Mach wave rather than vortices impingement. Evidences are raised up by visualization of instantaneous flowfields and comparison with two more cases with lower and higher convective Mach number. Components and features of compression waves in supersonic cavity flow are summarized. In supersonic cavity flows, the compression waves contain: (1) leading-edge shock waves; (2) trailing-edge shock waves; (3) Mach waves; (4) feedback acoustic waves; (5) front wall reflection n wave and bottom wall reflection wave. POD analysis in frequency domain is an efficient technique to analyze wave propagation. The compression wave propagations inside cavity are affected by the high-speed recirculation flows.

References

[1] Sheryl M. Grace. An overview of computational aeroacoustic techniques applied to cavity noise prediction. AIAA 2001-0510. January, 2001

[2] Tim Colonius. An overview of simulation, modeling, and active control of flow/acoustic resonance in open cavities. AIAA Paper 2001-0076, 2001

[3] Robert C. Murrayand Gregory S. Elliott. Characteristics of the Compressible Shear-layer over a Cavity. AIAA JOURNAL.Vol. 39, No. 5, May 2001

[4] Adela Ben-Yakar and Ronald K. Hanson. Cavity Flame-Holders for Ignition and Flame Stabilization in Scramjets: An Overview. Journal of Propulsion and Power. 17(4) 869-877. 2001

[5] Asbury, S. C., Gunther, C. L., and Hunter, C. A., Passive Cavity Concept for Improving the Off-Design Performance of Fixed-Geometry Exhaust Nozzles, AIAA Paper 96-2541, July 1996.

[6] Plentovich, E., "Three-dimensional Cavity Flow Fields at Subsonic and Transonic Speeds," Tech. Rep. TM 4209. NASA Langley Research Center, 1992.

[7] N. Zhuang, F. S. Alvi, M. B. Alkislar, and C. Shih. Supersonic Cavity Flows and Their Control. AIAA Journal. 44(9),2006: (2118-2128)

[8] N. Zhuang, F. S. Alvi and C. Shih. Another Look at Supersonic Cavity Flows and Their Control. AIAA Paper 2005-2803, 2001

[9] A. Powell. On the edge tone. Journal of the Acoustical Society of America. 33:395.1961.

[10] Rossiter, J. E.,"Wind-Tunnel Experiments on the Flowover Rectangular Cavities at Subsonic and Transonic Speeds," Aeronautical Research Council, Repts. and Memoranda No. 3438, London, Oct. 1964.

[11] Heller, H., Holmes, G., and Covert, E., "Flow-induced Pressure Oscillations in Shallow Cavities", AFFDL TR-70-104

[12] Ö. H. Ünalmis, N. T. Clemens, and D. S. Dolling. Experimental Study of Shear-Layer/Acoustics Coupling in Mach 5 Cavity Flow. AIAA Journal, 39(2), 2001: 242-252

[13] N. S. Vikramaditya and Job Kurian. Pressure Oscillations from Cavities with Ramp. AIAA Journal, 47(12), 2009: 2974-2984

[14] Hanno H. Heller and Donald B. Bliss. The Physical Mechanism of Flow-induced Pressure in Cavities and Concepts for Their Suppression. AIAA Paper 1975-491, 1975

[15] Rockwell, D., and Naudascher, E., “Review: Self-Sustaining Oscillations of Flow Past Cavities,” Journal of Fluids Engineering, 100(2), 1978, pp. 152–165.
[16] S. J. Lawson and G. N. Barakos. Assessment of Passive Flow Control for Transonic Cavity Flow Using Detached-Eddy Simulation. Journal of Aircraft, 46(3), 2009:1009-1029
[17] Chung-Jen Tam, Paul D. Orkwis and Peter J. Disimile. Supersonic open cavity flow physics ascertained from algebraic turbulence model simulations. AIAA Paper 96-0075. January 1996
[18] Shih, S. H., Hamed, A., and Yeuan, J. J., "Unsteady Supersonic Cavity Flow Simulations Using Coupled k-epsilon and Navier-Stokes Equations," AIAA Journal, Vol. 32, No. 10, October 1994, pp. 2015-2021.
[19] Zhang, X., "Compressible Cavity Flow Oscillation due to Shear-layer Instabilities and Pressure Feedback," AIAA Journal, Vol. 33, No. 8, August 1995, pp. 1404-1411.
[20] Selin Aradag, Doyle D. Knight. Simulation of Supersonic Flow over a Cavity. AIAA Paper 2005-848. January 2005
[21] Donald P. Rizzetta and Miguel R. Visbal Large-Eddy Simulation of Supersonic Cavity Flowfields Including Flow Control. AIAA Journal,41( )8, 2003:1452-1462
[22] A. Hamed, D. Basu, K. Das. Detached Eddy Simulations of Supersonic Flow over Cavity. AIAA 2003-549. January 2003
[23] Arunajatesan, S., Sinha, N., and Menon, S., "Towards Hybrid LES-RANS Computations of Cavity Flowfields," AIAA Paper, 2000-0401, 2000.
[24] Liang Jianhan, Sun Mingbo. LES of Cavity Oscillation Characteristics in High-Speed Flows. AIAA 2009-7231, 2009
[25] Taku Nonomura and Kozo Fujii. Effects of difference scheme type in high-order weighted compact nonlinear schemes. Journal of Computational Physics. 228 (2009) 3533‑3539
[26] Taku Nonomura, Nobuyuki Iizuka, Kozo Fujii. Freestream and vortex preservation properties of high-order WENO and WCNS on curvilinear grids. Computers and Fluids 39 (2010) 197‑214
[27] Taku Nonomura and Kozo Fujii. Computational Analysis of Characteristics of Mach Wave Sources in Supersonic Free-jets. AIAA 2009-16, 2009
[28] Shima, E. and Jounouchi, T., Role of CFD in Aeronautical Engineering (No. 14) -AUSM type Upwind Schemes, Proceedings of the 14th NAL Symposium on Aircraft Computational Aerodynamics, National Aeroautical Laboratory, Tokyo, 1997, pp. 7-12.
[29] Hiroyuki Nishida, Taku Nonomura. ADI-SGS scheme on ideal magnetohydrodynamics. Journal of Computational Physics. 228 (2009) 3182–3188
[30] D. A. Lyn and W. Rodi. The flapping shear layer formed by flow separation from the forward corner of a square cylinder. Journal of Fluid Mechanism, 267: 353-376, 1994
[31] Lionel Larchev êque, Pierre Sagaut, Ivan Mary and Odile Labb é. Large-eddy simulation of a compressible flow past a deep cavity. Physis of Fluids,15(1), 2003
[32] G. berkooz, P. Holmes, and J. L. Lumley. The Proper Orthogonal Decomposition in the Analysis of Turbulent Flows. Annual review of Fluid Mechanism, 25:539-575. 1993.
[33] Yang Qin and Fu Song. Analysis of flow structures in supersonic plane mixing layer using the POD method. Science in China series G: Physis, mechanics and Astronomy, 51(5): 541-558
[34] J. B. Freund and T. Colonius. Turbulence and sound-field POD analysis of a turbulent jet. Aeroacoustics, 8(4):337‑354, 2009.
[35] C.W. Rowley, T. Colonius, and R.M. Murray. POD based models on self-sustained oscillations in the flow past an open cavity. AIAA Paper 2000-1969, 2000.